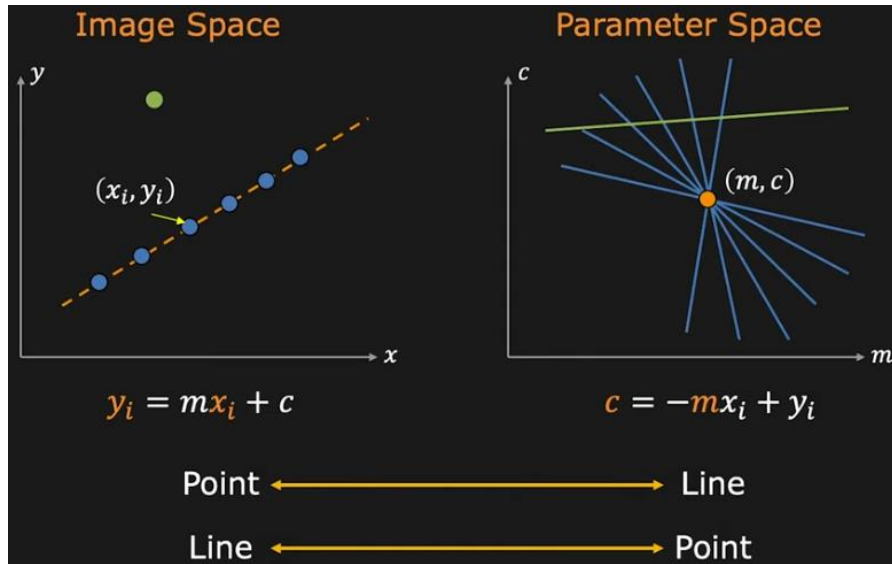


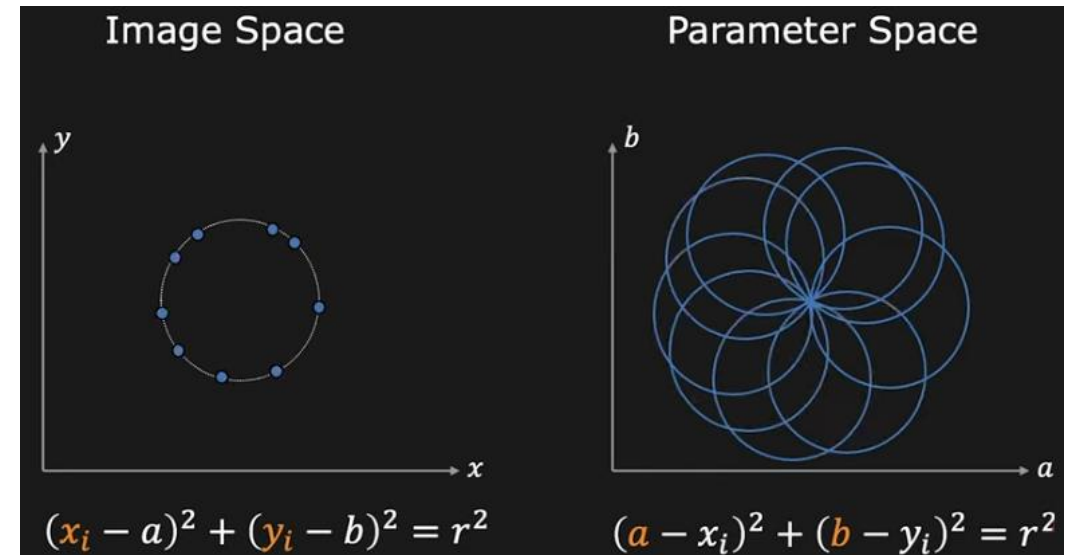
Hough Transform

- We know how to detect:

Straight lines

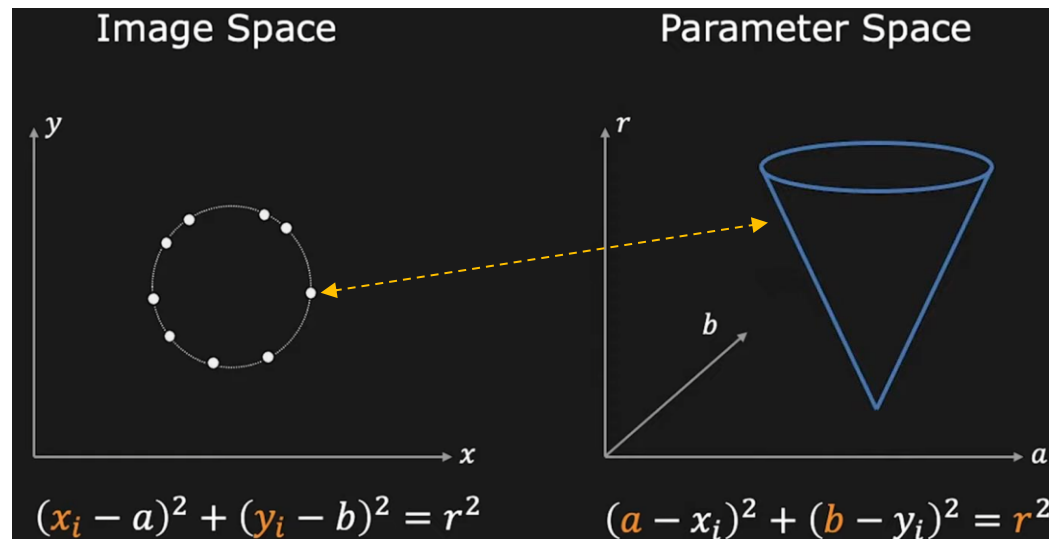


Circles (r known)

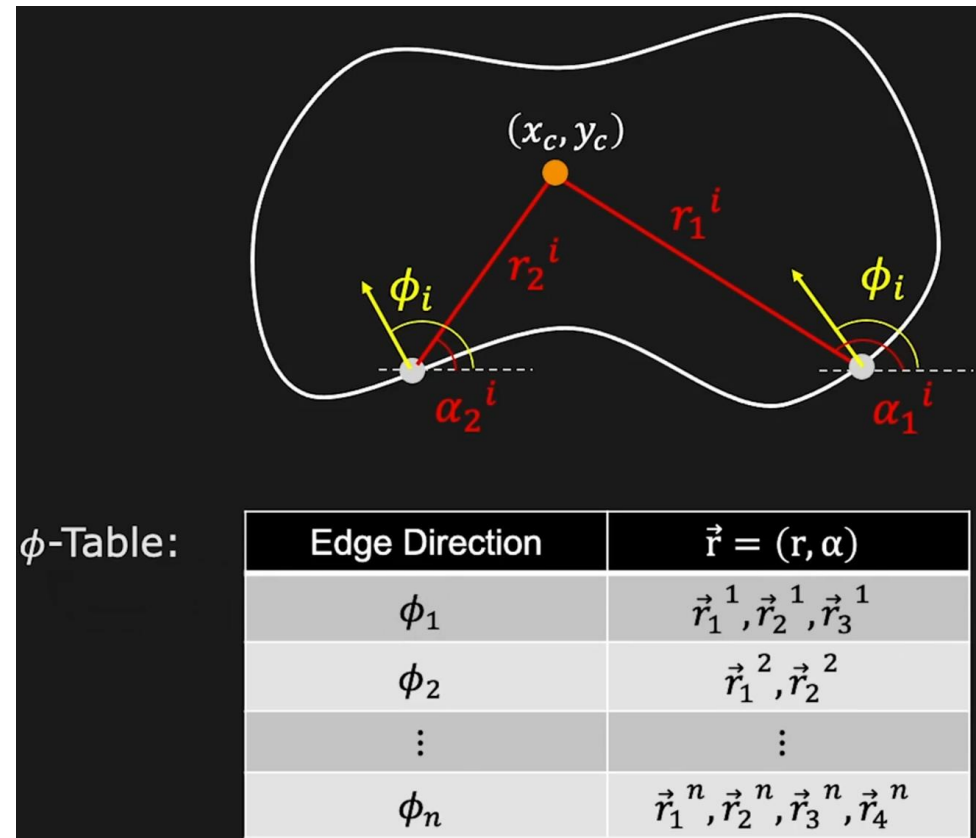


Hough Transform

- We know how to detect:
Circles (r unknown)

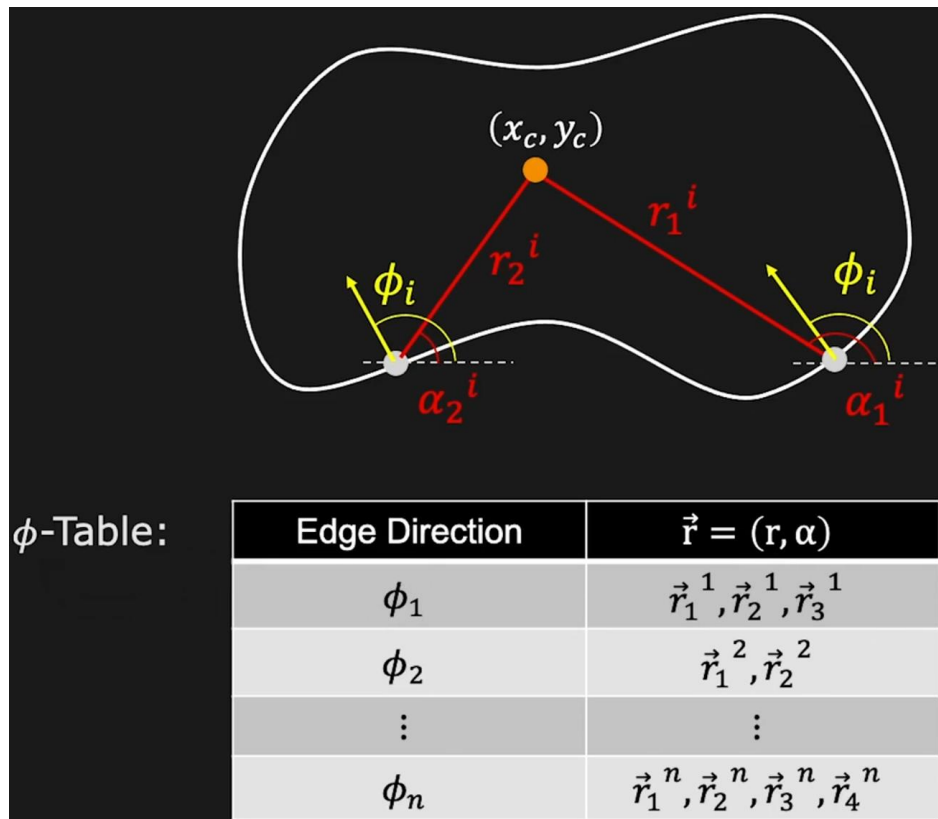


Arbitrary shapes
(parameterized following the
Generalized Hough Transform)



Hough Transform

- In the context of the Generalized Hough Transform,
 - what happens if we want to deal as well with rotation and scale?? **We know that our accumulator increases dimensionality, what should we also change our ϕ -Table?**



- Create **accumulator array** $A(x_c, y_c)$
- Set $A(x_c, y_c) = 0$ for all (x_c, y_c)
- For each edge point (x_i, y_i, ϕ_i) ,

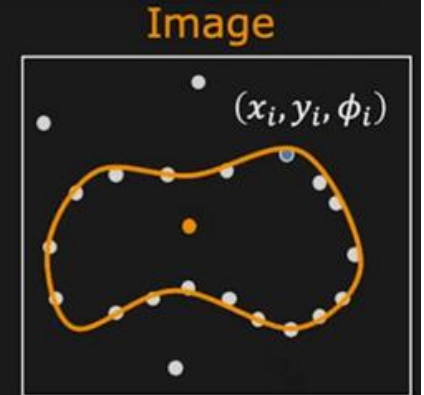
For each entry $\phi_i \rightarrow \vec{r}_k^i$ in ϕ -table,

$$x_c = x_i \pm r_k^i \cos(\alpha_k^i)$$

$$y_c = y_i \pm r_k^i \sin(\alpha_k^i)$$

$$A(x_c, y_c) = A(x_c, y_c) + 1$$

- Find local maxima in $A(x_c, y_c)$



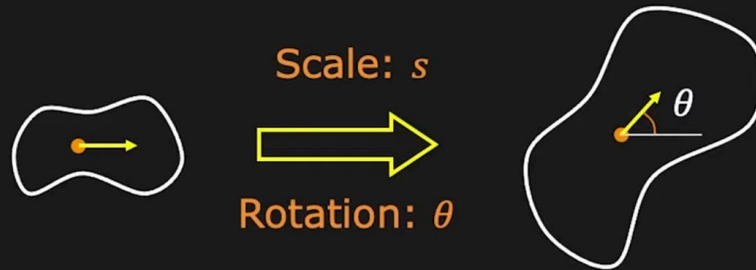
$A(x_c, y_c)$

x_c	0	0	0	0	0
0	0	2	0	1	0
0	0	0	4	1	0
0	2	0	0	0	0
0	0	0	0	1	0
y_c					

Hough Transform

- In the context of the generalized Hough Transform,
 - We know that our accumulator increases dimensionality, what should we also change our ϕ -Table?

Handling Scale And Rotation



Use Accumulation Array: $A(x_c, y_c, s, \theta)$

$$x_c = x_i \pm r_k^i \cdot s \cos(\alpha_k^i + \theta)$$

$$y_c = y_i \pm r_k^i \cdot s \sin(\alpha_k^i + \theta)$$

$$A(x_c, y_c, s, \theta) = A(x_c, y_c, s, \theta) + 1$$

Huge Memory and Computationally Expensive!

The ϕ -Table doesn't change!

The space of possible scales and rotations is sampled to compute x_c and y_c .

Hough Transform

- In the context of the generalized Hough Transform,
 - **We know that our accumulator increases dimensionality, what should we also change our ϕ -Table?**

```
for each edge point (xi, yi,  $\phi_i$ ) in the image:
    for each scale s:
        for each rotation  $\theta$ :
             $\phi_{\text{model}} = \phi_i - \theta$ 
            for each entry (rk,  $\alpha_k$ ) in R_table[ $\phi_{\text{model}}$ ]:
                 $x_c = x_i + r_k * s * \cos(\alpha_k + \theta)$ 
                 $y_c = y_i + r_k * s * \sin(\alpha_k + \theta)$ 
                if (xc, yc) inside bounds:
                     $A[x_c, y_c, s, \theta] += 1$ 
```

The ϕ -Table doesn't change!

The space of possible scales and rotations is sampled to compute x_c and y_c .